

cable), first enable the camera interface in Raspberry Pi config setting. Then install Edge Impulse on Raspberry Pi by running the following commands on terminal:

```
• curl -sL https://deb.nodesource.com/setup_12.x | sudo bash -
• sudo apt install -y gcc g++ +
make build-essential nodejs sox
gststreamer1.0-tools gstreamer1.0-
```

plugins-good gstreamer1.0-plugins-base gstreamer1.0-plugins-base-apps

```
• sudo npm install edge-impulse-linux -g --unsafe-perm
```

After setting up Edge Impulse, create and train the ML model. So, open the Edge Impulse website and then enter your name and email id and sign up. Create a new project in Edge Impulse, open Linux terminal

and run the command *edge-impulse-linux*

When you select the project, your device will try to connect with Edge Impulse. If the connection is proper, the device detection of the Edge Impulse Raspberry Pi cam will be shown, and you will get a link. Open the link in a web browser and get the data acquisition interface to take a photo or any other data for ML mode. Photos can be taken of objects like a bottle or cup, and any other item (including human beings) for the ML model.

For data acquisition, take at least 100 photos of the different objects that you want to classify and use for training and testing the ML model. To rebalance the data, the ratio of splitting should be 70:30.

Go to dashboard, where the Labeling Method should be bounding boxes (object detection). Label all the objects and see this in Labeling Queue. Then go to Impulse Design, where the image width and image height should be 320*320. You may change the object detection pro-

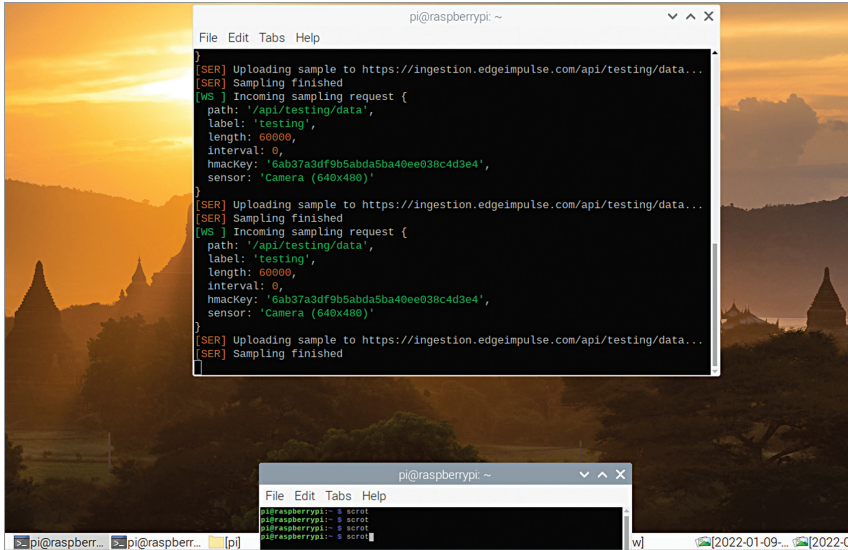


Fig. 3: The terminal

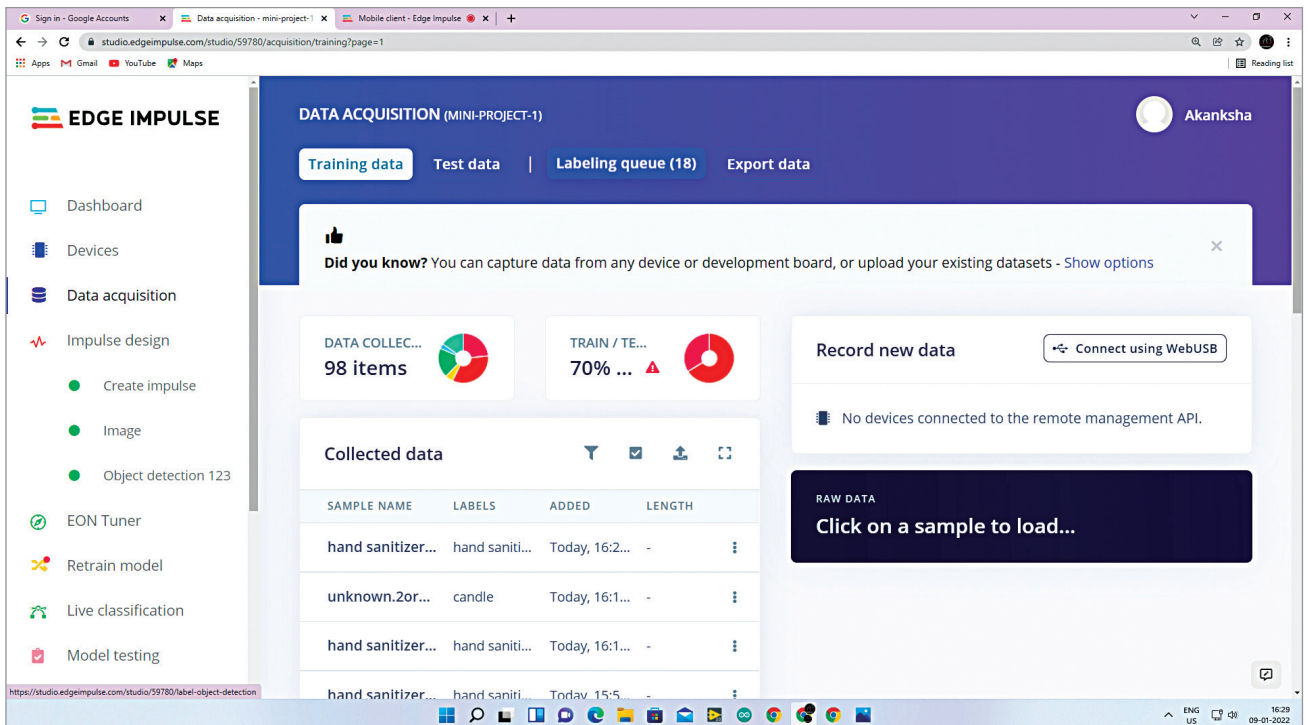


Fig. 4: Data acquisition